

## QUESTIONS

- Q1 Student Management System (BL2 - Apply) 10 marks
- Q2 Library Book Management (BL2 - Apply) 10 marks
- Q3 Employee Salary Calculation (BL2 - Apply) 10 marks
- Q4 Shape Area Calculator using Polymorphism (BL2 - Apply) 10 marks
- Q5 Vehicle Registration System (BL2 - Apply) 10 marks
- Q6 Bank Account Hierarchy (BL4 - Analyze) 10 marks
- Q7 Hospital Patient Record System (BL4 - Analyze) 10 marks
- Q8 E-Commerce Product Catalog (BL4 - Analyze) 10 marks
- Q9 University Staff Hierarchy (BL4 - Analyze) 10 marks
- Q10 Smart Home Device Controller (BL4 - Analyze) 10 marks

10/10 answered

## 9. University Staff Hierarchy (BL4 - Analyze)

10 marks

Analyze the design of a Staff base class extended by TeachingStaff and NonTeachingStaff. Further extend TeachingStaff into Professor and Lecturer (multilevel inheritance). For each level:

- Identify which attributes should be private vs protected
- Override calculateAllowance() at each level
- Analyze how protected access enables inheritance without full exposure. Justify the use of multilevel inheritance and explain how access modifiers enforce data hiding.

OOP Concepts: Multilevel Inheritance | Access Modifiers | Data Hiding

## Your Answer

```
START
CLASS Staff
    PRIVATE staffid, name
    PROTECTED salary
    METHOD calculateAllowance()
        DISPLAY 0
    END METHOD
END CLASS
CLASS TeachingStaff EXTENDS Staff
    METHOD calculateAllowance()
        DISPLAY salary * 0.20
    END METHOD
END CLASS
CLASS Professor EXTENDS TeachingStaff
    METHOD calculateAllowance()
        DISPLAY salary * 0.30
    END METHOD
END CLASS
CLASS Lecturer EXTENDS TeachingStaff
    METHOD calculateAllowance()
        DISPLAY salary * 0.25
    END METHOD
END CLASS
CLASS NonTeachingStaff EXTENDS Staff
    METHOD calculateAllowance()
        DISPLAY salary * 0.15
    END METHOD
END CLASS
CREATE Professor p
CREATE Lecturer l
CREATE NonTeachingStaff n
CALL p.calculateAllowance()
CALL l.calculateAllowance()
CALL n.calculateAllowance()
STOP
```

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10/10 answered

## 8. E-Commerce Product Catalog (BL4 - Analyze)

10 marks

Analyze and design a Product class hierarchy for an e-commerce system:

- Electronics — with warranty, brand, voltage
- Clothing — with size, fabric, gender
- Grocery — with expiryDate and weight

Override an applyDiscount() method for each category with different discount rules. Analyze why polymorphism makes the cart checkout process flexible and extensible. OOP Concepts: Polymorphism | Inheritance | Method Overriding

## Your Answer

```
START
CLASS Product
    name, price
    METHOD applyDiscount()
        DISPLAY price
    END METHOD
END CLASS
CLASS Electronics EXTENDS Product
    warranty, brand, voltage
    METHOD applyDiscount()
        price = price - (price * 10 / 100)
        DISPLAY price
    END METHOD
END CLASS
CLASS Clothing EXTENDS Product
    size, fabric, gender
    METHOD applyDiscount()
        price = price - (price * 20 / 100)
        DISPLAY price
    END METHOD
END CLASS
CLASS Grocery EXTENDS Product
    expiryDate, weight
    METHOD applyDiscount()
        price = price - (price * 5 / 100)
        DISPLAY price
    END METHOD
END CLASS
CREATE Electronics e
CREATE Clothing c
CREATE Grocery g
CALL e.applyDiscount()
CALL c.applyDiscount()
CALL g.applyDiscount()
STOP
```



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Back

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Q8 E-commerce Product Catalog (BL4 - Analyze) 10 marks

Q9 University BOP Identity (BL4 - Analyze) 10 marks

Q10 Hospital Ward Details (BL4 - Apply) 10 marks

10/10 answered

**7. Hospital Patient Record System [BL4 - Analyze]** 10 marks

Design a Patient base class and analyze how it extends into:

- Inpatient — with ward number, admission date, and daily charges
- Outpatient — with appointment date and consultation fee

Analyze encapsulation requirements for sensitive data (diagnosis, prescriptions). Override a generateBill() method for each type and compare the billing logic differences. OOP Concepts: Encapsulation | Inheritance | Polymorphism

**Your Answer**

```
START
CLASS BankAccount
    attribute: balance
    METHOD withdraw(amount)
        balance = balance - amount
    END METHOD
END CLASS
CLASS SavingsAccount EXTENDS BankAccount
    METHOD withdraw(amount)
        IF amount <= balance THEN
            balance = balance - amount
        ELSE
            DISPLAY "Insufficient Balance"
        END IF
    END METHOD
END CLASS
CLASS CurrentAccount EXTENDS BankAccount
    overdraftLimit
    METHOD withdraw(amount)
        IF amount <= balance + overdraftLimit THEN
            balance = balance - amount
        ELSE
            DISPLAY "Overdraft Limit Exceeded"
        END IF
    END METHOD
END CLASS
CREATE SavingsAccount s
CREATE CurrentAccount c
CALL s.withdraw(500)
CALL c.withdraw(500)
STOP
```

Save Answer

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10/10 answered

**6. Bank Account Hierarchy [BL4 - Analyze]** 10 marks

Develop pseudocode for a BankAccount superclass and two subclasses — SavingsAccount (with interest calculation) and CurrentAccount (with overdraft facility). Analyze and implement:

- Inheritance of common attributes (account, balance, holder)
- Method overriding for withdrawal behavior
- How overdraft protection differs from savings limit

Compare the behavior of withdrawing across both subclasses and justify the design choices. OOP Concepts: Inheritance | Method Overriding | Polymorphism

**Your Answer**

```
START
CLASS BankAccount
    attribute: holder, balance
    METHOD withdraw(amount)
        balance = balance - amount
    END METHOD
END CLASS
CLASS SavingsAccount EXTENDS BankAccount
    METHOD withdraw(amount)
        IF amount <= balance THEN
            balance = balance - amount
        ELSE
            DISPLAY "Insufficient Balance"
        END IF
    END METHOD
END CLASS
CLASS CurrentAccount EXTENDS BankAccount
    overdraftLimit
    METHOD withdraw(amount)
        IF amount <= balance + overdraftLimit THEN
            balance = balance - amount
        ELSE
            DISPLAY "Overdraft Limit Exceeded"
        END IF
    END METHOD
END CLASS
CREATE SavingsAccount s
CREATE CurrentAccount c
CALL s.withdraw(500)
CALL c.withdraw(500)
STOP
```

Save Answer



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10/10 answered

**5. Vehicle Registration System [BL3 - Apply]** 10 marks

Design a Vehicle class with attributes (regNo, owner, fuelType). Extend it into:

- TwoWheeler — with engine capacity
- FourWheeler — with seating capacity and AC status

Apply inheritance to reuse common attributes and override a displayDetails() method in each subclass. OOP Concepts: Inheritance | Method Overriding | Subclass

**Your Answer**

```
START
CLASS Vehicle
    regNo, owner, fuelType
    METHOD displayDetails()
        DISPLAY regNo, owner, fuelType
    END METHOD
END CLASS
CLASS TwoWheeler EXTENDS Vehicle
    engineCapacity
    METHOD displayDetails()
        DISPLAY regNo, owner, fuelType, engineCapacity
    END METHOD
END CLASS
CLASS FourWheeler EXTENDS Vehicle
    seatingCapacity, ACstatus
    METHOD displayDetails()
        DISPLAY regNo, owner, fuelType, seatingCapacity, ACstatus
    END METHOD
END CLASS
CREATE TwoWheeler t
CREATE FourWheeler f
CALL t.displayDetails()
CALL f.displayDetails()
STOP
```



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Q7 7. Hospital Patient Record System (BL4 - Analyze) 10 marks

Q8 8. E-Commerce Product Catalog (BL4 - Analyze) 10 marks

Q9 9. University Staff Hierarchy (BL4 - Analyze) 10 marks

Q10 10. Smart Meter Data Controller (BL4 - Analyze) 10 marks

10/10 answered

Shape Area Calculator using Polymorphism (BL3 - Apply) 10 marks

Create a Shape base class with an overridable calculateArea() method. Apply this to:

- Rectangle — use using length and breadth
- Triangle — use using base and height

Use method overriding to demonstrate runtime polymorphism. Call calculateArea() for each object.  
OOP Concepts: Inheritance | Polymorphism | Method Overriding

Your Answer

```
START
CLASS Shape
METHOD calculateArea()
    DISPLAY "Area"
END METHOD
END CLASS
CLASS Circle EXTENDS Shape
    radius
    METHOD calculateArea()
        DISPLAY 3.14 * radius * radius
    END METHOD
END CLASS
CLASS Rectangle EXTENDS Shape
    length, breadth
    METHOD calculateArea()
        DISPLAY length * breadth
    END METHOD
END CLASS
CLASS Triangle EXTENDS Shape
    base, height
    METHOD calculateArea()
        DISPLAY 0.5 * base * height
    END METHOD
END CLASS
CREATE Circle c
CREATE Rectangle r
CREATE Triangle t
CALL c.calculateArea()
CALL r.calculateArea()
CALL t.calculateArea()
STOP
```

Save Answer



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10/10 answered

Employee Salary Calculation (BL3 - Apply) 10 marks

Write pseudocode for an Employee class with encapsulated attributes (empId, name, basicSalary, designation). Implement methods to:

- Calculate gross salary (basic + HRA + DA)
- Calculate net salary after deductions (PF, tax)
- Display pay slip

Apply data hiding so salary components are not directly accessible.  
OOP Concepts: Encapsulation | Data Hiding | Methods

Your Answer

```
START
CLASS Book
PRIVATE title, author, ISBN, isAvailable
METHOD setDetails(a, b, c)
    title = a
    author = b
    ISBN = c
    isAvailable = TRUE
END METHOD
METHOD isAvailable()
    isAvailable = FALSE
END METHOD
METHOD returnBook()
    isAvailable = TRUE
END METHOD
METHOD getDetails()
    DISPLAY title, author, ISBN, isAvailable
END METHOD
END CLASS
CREATE Book b1, b2, b3
CALL b1.setDetails("Java", "James", "101")
CALL b2.setDetails("Python", "Guido", "102")
CALL b3.setDetails("C++", "Bjarne", "103")
CALL b1.isAvailable()
CALL b2.isAvailable()
CALL b1.returnBook()
CALL b1.getDetails()
CALL b2.getDetails()
CALL b3.getDetails()
STOP
```

Save Answer



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10/10 answered

Library Book Management (BL3 - Apply) 10 marks

Design a Book class with private attributes (title, author, ISBN, isAvailable). Implement methods to:

- Issue a book (change availability status)
- Return a book
- Display book details using getter methods

Demonstrate object creation for at least 3 books and simulate issue/return operations.  
OOP Concepts: Encapsulation | Getters & Setters | Objects

Your Answer

```
START
CLASS Book
PRIVATE title, author, ISBN, isAvailable
METHOD setDetails(a, b, c)
    title = a
    author = b
    ISBN = c
    isAvailable = TRUE
END METHOD
METHOD issueBook()
    isAvailable = FALSE
END METHOD
METHOD returnBook()
    isAvailable = TRUE
END METHOD
METHOD getDetails()
    DISPLAY title, author, ISBN, isAvailable
END METHOD
END CLASS
CREATE Book b1, b2, b3
CALL b1.setDetails("Java", "James", "101")
CALL b2.setDetails("Python", "Guido", "102")
CALL b3.setDetails("C++", "Bjarne", "103")
CALL b1.issueBook()
CALL b2.issueBook()
CALL b1.returnBook()
CALL b1.getDetails()
```

Save Answer

## QUESTIONS

Q1  
Student Management System  
(BL3 – Apply)  
10 marks

Q2  
Library Book Management (BL3  
– Apply)  
10 marks

Q3  
Employee Salary Calculation  
(BL3 – Apply)  
10 marks

Q4  
Shape Area Calculator using  
Polymorphism (BL3 – Apply)  
10 marks

Q5  
5. Vehicle Registration System  
(BL3 – Apply)  
10 marks

Q6  
6. Bank Account Hierarchy (BL4  
– Analyze)  
10 marks

Q7  
7. Hospital Patient Record  
System (BL4 – Analyze)  
10 marks

Q8  
8. E-Commerce Product  
Catalog (BL4 – Analyze)  
10 marks

## Q1 Student Management System (BL3 – Apply)

10 marks

Write pseudocode to design a Student class with attributes (name, rollNo, marks) and methods to:

- Accept student details
- Calculate average marks
- Display result (Pass/Fail)

Apply encapsulation and data hiding by restricting direct access to attributes.  
OOP Concepts: Encapsulation | Data Hiding | Classes & Objects

## Your Answer

```
START
CLASS Student
PRIVATE name, rollNo, marks, average
METHOD acceptDetails()
INPUT name, rollNo, marks
END METHOD
METHOD calculateAverage()
average ← marks
END METHOD
METHOD displayResult()
IF average >= 40 THEN
    DISPLAY "PASS"
ELSE
    DISPLAY "FAIL"
END IF
END METHOD
END CLASS
CREATE Student object s
CALL s.acceptDetails()
CALL s.calculateAverage()
CALL s.displayResult()
STOP
```

## QUESTIONS

Q1  
Student Management System  
(BL3 – Apply)  
10 marks

Q2  
Library Book Management (BL3  
– Apply)  
10 marks

Q3  
Employee Salary Calculation  
(BL3 – Apply)  
10 marks

Q4  
Shape Area Calculator using  
Polymorphism (BL3 – Apply)  
10 marks

Q5  
5. Vehicle Registration System  
(BL3 – Apply)  
10 marks

Q6  
6. Bank Account Hierarchy (BL4  
– Analyze)  
10 marks

Q7  
7. Hospital Patient Record  
System (BL4 – Analyze)  
10 marks

Q8  
8. E-Commerce Product  
Catalog (BL4 – Analyze)  
10 marks

Q9  
9. University Staff Hierarchy  
(BL4 – Analyze)  
10 marks

Q10  
10. Smart Home Device  
Controller (BL4 – Analyze)  
10 marks

10/10 answered

## Q10 10. Smart Home Device Controller (BL4 – Analyze)

10 marks

Analyze and design a SmartDevice superclass with subclasses: SmartLight, SmartThermostat, and SmartLock. Override an executeCommand(cmd) method in each, where the same command (e.g., 'on', 'off', 'status') produces device-specific behavior. Analyze:

- How polymorphism allows a single controller loop to manage all devices
- Security implications of data hiding in SmartLock (PIN must never be exposed) Design the class hierarchy and explain the OOP principles applied at each level.

OOP Concepts: Polymorphism | Encapsulation | Inheritance

## Your Answer

```
START
CLASS SmartDevice
    deviceName
    METHOD executeCommand(cmd)
        DISPLAY "Command Executed"
    END METHOD
END CLASS
CLASS SmartLight EXTENDS SmartDevice
    METHOD executeCommand(cmd)
        DISPLAY "Light ", cmd
    END METHOD
END CLASS
CLASS SmartThermostat EXTENDS SmartDevice
    METHOD executeCommand(cmd)
        DISPLAY "Thermostat ", cmd
    END METHOD
END CLASS
CLASS SmartLock EXTENDS SmartDevice
    PRIVATE PIN
    METHOD executeCommand(cmd)
        DISPLAY "Lock ", cmd
    END METHOD
END CLASS
CREATE SmartLight l
CREATE SmartThermostat t
CREATE SmartLock s
CALL l.executeCommand("ON")
CALL l.executeCommand("STATUS")
CALL s.executeCommand("OFF")
STOP
```

Save Answer